Class: Algorithms & Data Structures Semester: Spring 2025

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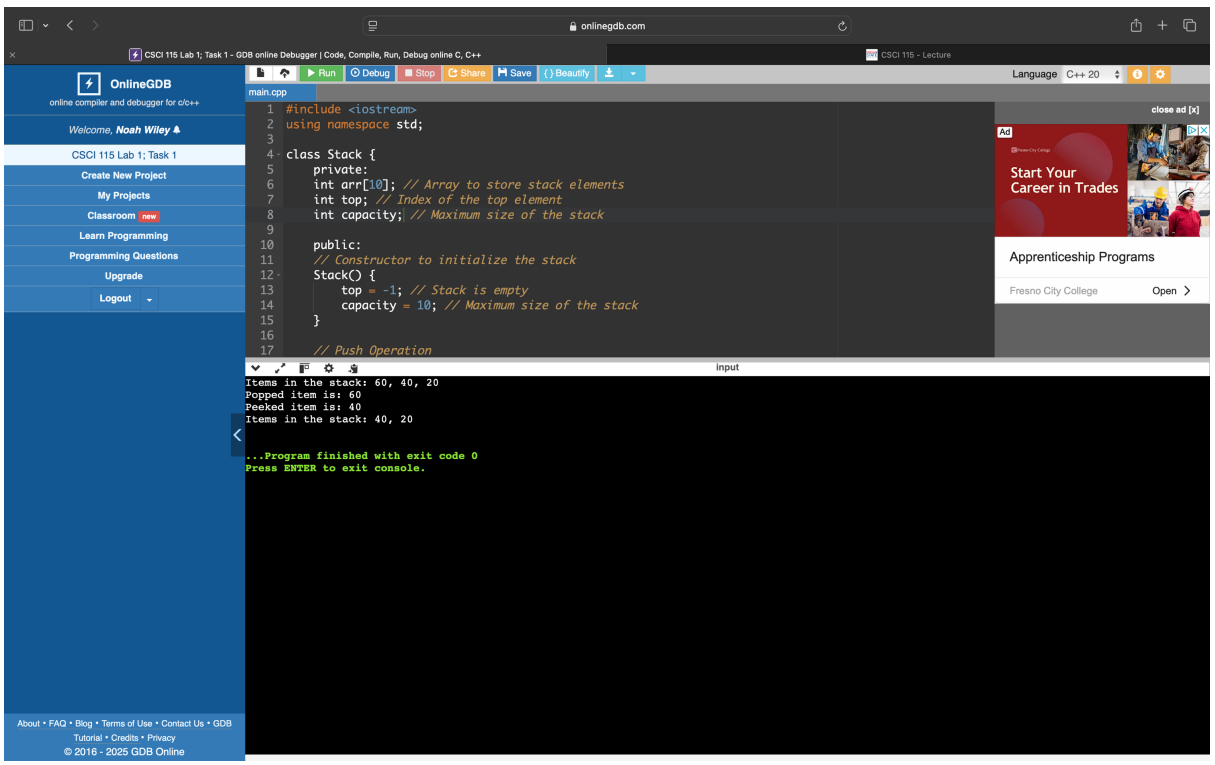
Lab Number - Section 09

CSCI 115 Lab 1 Report; Task 1

- Statement of Objectives:
  - This lab experiment focuses on understanding and implementing the fundamental concepts of stacks, specifically their operations. The primary objective is to review and solidify our knowledge of stacks as a data structure, implementing it without relying on built-in classes. We aim to demonstrate the push, pop, and peek operations, exploring their practical utility and behavior through C++ programming. This report outlines the experimental procedure, analysis, challenges faced, and conclusions from this lab.

- Experimental Procedure:
  - To achieve the lab objectives, the following procedure was followed:
    - Define the Stack Data Structure: Implemented a stack using an array with a fixed size of 10
    - Implement Operations: Developed functions to support the push, pop, and peek operations
      - Push: Added elements to the top of the stack, ensuring no overflow occurred
      - Pop: Removed and returned the top-most element of the stack, ensuring no underflow occurred
      - Peek: Retrieved the top-most element without removing it
    - Perform Operations: Executed the following tasks step-by-step:
      - Pushed integers 20, 40, and 60 onto the stack
      - Printed the current stack contents
      - Popped the top-most item and printed it
      - Peeked the stack and printed the top-most item
      - Printed the final stack contents after performing these operations
    - Validate Outputs: Ensured the results matched the expected output specified in the lab assignment.

- Analysis
  - The following results were obtained from the experiment:
    - Initial Stack State: After pushing 20, 40, and 60, the stack contents were printed as: 60, 40, 20
    - Popped Item: The top-most item, 60, was successfully removed and printed: Popped item is: 60
    - Peeked Item: The next top-most item, 40, was retrieved without removal: Peaked item is: 40
    - Final Stack State: After the pop operation, the remaining stack contents were: 40, 20



- Encountered Problems:
  - The only error I encountered is trying to push more elements than the array size allowed. This was resolved by adding a condition to check for overflow before pushing an element

- Conclusions:
  - Through this lab, the following concepts were reinforced:
    - The implementation and manipulation of stacks using arrays
    - The behavior of push, pop, and peek operations, along with edge case handling
    - The importance of error handling in preventing overflow and underflow in stack operations
    - Debugging techniques to address runtime errors and logical discrepancies
      - This exercise provided a hands-on understanding of stack operations and their practical implications in data structure management.

- References
  - Lecture Slides on Stacks and Linked Lists